`std::condition_variable` instance are serialized. A call to `notify_one()` or `notify_all()` will only wake threads that started waiting *prior* to that call.

Std::Condition_Variable::Wait_For Member Function Overload That T- Takes a Predicate

Wait until the `std::condition_variable` is woken by a call to `notify_one()` or `notify_all()` and the predicate is `true`, or until the specified time period has elapsed.

Declaration

```
template<typename Rep,typename Period,typename Predicate>
bool wait_for(
    std::unique_lock<std::mutex>& lock,
    std::chrono::duration<Rep,Period> const& relative_time,
    Predicate pred);
```

Preconditions

The expression `pred()` shall be valid and shall return a value that's convertible to `bool`. `lock.owns_lock()` shall be `true`, and the lock shall be owned by the thread calling `wait()`.

Effects

As-if

```
internal_clock::time_point end=internal_clock::now()+relative_time;
while(!pred())
{
    std::chrono::duration<Rep,Period> remaining_time=
        end-internal_clock::now();
    if(wait_for(lock,remaining_time)==std::cv_status::timeout)
        return pred();
}
return true;
```

Returns